

P SANGAM

AI Software Engineer

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SUMMARY

AI Engineer with hands-on experience building and integrating machine learning, computer vision, and generative AI components into working, business-facing systems, spanning IoT sensor pipelines, embedded hardware integration, forecasting models, and LLM-powered advisory tools. Comfortable owning problems end-to-end in a fast-moving, resource-constrained environment with minimal senior technical oversight. Skilled at translating technical trade-offs into terms business stakeholders and leadership can act on. Specializing in production Generative AI engineering, agentic workflow architecture, RAG evaluation, and real-time hardware-software integration.

CORE SKILLS

AI & ML: Machine Learning, Deep Learning, Computer Vision, Time-Series Forecasting, LightGBM, LSTM, OpenCV, YOLOv8.

AI & Agentic Systems: Agentic Workflows (langchain/ lanGraph/ CrewAI), Tool calling / Function calling, RAG Systems, Vector Databases (FAISS), Prompt Engineering & Evaluation.

Tools Environments: Cursor, Claude API, Antigravity, Hermes, Docker, FastAPI, Git, CI/CD (Foundation).

Consultative & Delivery: Client Technical Demos, Requirements Gathering, Agile/Scrum, Stakeholder Communication.

Software & Systems: Python, ReactJS, Typescript, SQL, GraphQL(foundational).

EXPERIENCE

EAGE Technologies India Pvt. Ltd.

Bengaluru, KA

AI Software Engineer

Feb 2026 – Present

- Built and benchmarked LightGBM and LSTM forecasting models for solar power output using 12 years of location-based historical data, reaching evaluation scores of 89% and 80% respectively, designed retraining triggers and performance-ratio-based drift monitoring.
- Architected an agentic decision pipeline integrating LightGBM/LSTM forecasts with rule-based risk logic and LLM function calling, generating real-time operational directives and automated risk alerts for solar assets.
- Built a real-time computer vision pipeline (YOLOv8, InsightFace, FAISS) for face detection, distance estimation, and object detection, integrated into an STM32 hardware-software pipeline for a client defense application.
- Designed an IoT-to-cloud Agri-Soil Intelligence engine **under zero-historical-data constraints**, engineering cold-start threshold logic and location-based climate API pipelines for 2 pilot locations.
- Interfaced directly with client stakeholders and engineering leads to translate complex operational needs into technical specifications and deploying pilot systems in live customer environments.

Research & Development Intern

Jul 2025 – Jan 2026

- Conducted applied research and development of machine learning and deep learning solutions across predictive analytics, computer vision, and forecasting use cases.
- Evaluated model performance through experimentation and data analysis to guide solution design decisions.

Code Nimbus Solutions Pvt. Ltd.

Bengaluru, KA

AI & Frontend Intern

Sep 2024 – Jun 2025

- Built AI-powered facial recognition pipelines using deep learning and embedding extraction techniques.
- Developed user interfaces for AI applications, enabling visualization of model outputs and operational workflows.

PROJECTS

Solar Intelligence Platform: Forecasting and decision-support platform combining LightGBM and LSTM models (89% / 80% evaluation scores) trained on 12 years of historical data with rule-based risk analysis and an LLM-generated advisory layer, designed with retraining triggers and drift-detection monitoring.

Soil Intelligence Platform: IoT-to-cloud agricultural advisory system using GSM-based sensor data and location/climate APIs, threshold-based rule engine for soil health, crop recommendation, and disease risk across two pilot locations, built in the absence of historical data.

Defense Surveillance: Real-time computer vision system integrating pretrained YOLOv8, InsightFace, and FAISS for face detection, distance estimation, and object detection (including IR images), deployed on STM32 embedded hardware as part of a hardware-software integration pipeline.

EDUCATION

SDM Institute of Technology

B.E. in Artificial Intelligence & Data Science · 2021 – 2025 · CGPA: 7.69

Ujire, Dakshina Kannada, KA, India